ADVANCED UNIT · STUDENT WORKBOOK

ADVANCED WATER SAFETY

Health & Physical Education · Year 9–10 · Victorian Curriculum F–10 Version 2.0

This advanced unit builds on the Year 7–8 foundation. You will **analyse** Victorian and international drowning data, **evaluate** prevention strategies, **investigate** the physiology of drowning, and produce a **policy-level communication product** for a real audience.

STUDENT NAME _____

CLASS _____

TEACHER _____

Curriculum 2.0: VC2HP10P02 · VC2HP10P08 · VC2HP10P09 · VC2HP10M01

Focus Areas: Safety (S) · Health Literacy · Community Health Promotion

Welcome to Advanced Water Safety. You already know the fundamentals — flags, rip currents, DRS ABCD, and SwimSafe rules from Year 7–8. This unit asks you to go deeper: *why* do drowning rates remain so high despite awareness campaigns? Who is most at risk and why? What does the research say about what actually works? And what would *you* recommend to reduce Victorian drowning deaths?

1 Advanced Data Analysis — Victorian & Global Drowning Trends

KEY KNOWLEDGE — ADVANCED DATA CONTEXT

- Victoria's 52 drowning deaths in 2024–25 were 9% above the ten-year average of 48
- Non-fatal drownings were 60% above the 10-year average

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- People from multicultural backgrounds accounted for 37% of Victorian drowning deaths since 2015–16 (185 deaths); country of birth unknown in 22% of cases. Among 15–24-year-olds the share is higher again at 46% over the past five years (LSV Aquatic Injury Prevention Agenda 2026–27)
- Globally, around 300,000 people drown annually; 92% in low- and middle-income countries (WHO 2024)
- Low- and middle-income countries account for 92% of global drowning deaths (WHO, around 300,000 deaths a year)
- Males account for approximately 80% of global drowning deaths
- Children under 5 are the highest-risk age group globally
- In Australia, 1.2 people per 100,000 drown annually (RLSSA national data)

Sources: LSV — lsv.com.au/what-we-do/research-advocacy · WHO — who.int/news-room/fact-sheets/detail/drowning · RLSSA — royallifesaving.com.au/research · CDC — cdc.gov/drowning/prevention

Activity 1.1 — Comparative Data Analysis

Individual / pair · ~20 min

Use the data sources above and your workbook to complete the following analysis tasks. For each, show your working or cite your source.

Question 1.1a — Calculation

3 marks

Victoria's population in 2024–25 was approximately 6.9 million. Calculate Victoria's drowning rate per 100,000 population, then compare it to the national Australian rate (1.13 per 100,000, 2025/26). What does the difference tell you?

Calculation:

Victorian rate: _____ per 100,000 · National rate: 1.13 per 100,000 (2025/26)

What this tells us:

Question 1.1b — Trend Analysis

4 marks

Visit: **lsv.com.au/what-we-do/research-advocacy** — access the two most recent annual reports. Identify ONE trend that has worsened over that period and ONE that has improved. For each, suggest a possible reason for the trend.

Trend Direction	What has changed?	Possible reason

Worsening trend	*Describe the change...*	*Your explanation...*
Improving trend	*Describe the change...*	*Your explanation...*

Question 1.1c — Global Comparison `3 marks`

The WHO reports around 300,000 drowning deaths globally per year. Using this data and Victorian data, answer: If Victoria's drowning rate was at the global average for high-income countries (approximately 0.8 per 100,000), how many Victorian deaths per year would that represent? Show your calculation and explain what interventions might close this gap.

Calculation: _____

Expected deaths at 0.8 per 100,000: _____

Interventions that could close the gap: _____

_____

📊 **Data Visualisation Task**

In the space below (or on a separate page), create a bar graph or pie chart comparing Victorian drowning deaths by: (a) location type (beach, river/dam, pool), or (b) age group. Label axes, include a title, and write a 2-sentence interpretation below your graph.

Draw or attach your graph here...

Interpretation: _____

_____

2 Physiology of Drowning — What Happens in the Body

KEY KNOWLEDGE — PHYSIOLOGICAL PROCESSES

- **Hypoxia:** Oxygen deprivation. Brain cells begin dying within 4–6 minutes of oxygen loss. After 10 minutes without CPR, the chance of survival without significant brain damage is very low.
- **Cold water shock (1-10-1 rule):** ~1 min to control breathing; ~10 min of useful swimming; up to 1 hr before hypothermia incapacitates (varies greatly by water temperature, body composition, and clothing).
- **Wet vs dry drowning:** In most drowning cases (wet), water enters the lungs. In ~10–15% (dry), laryngospasm prevents water entry but still causes asphyxia.
- **Secondary drowning:** Rare but real — symptoms (coughing, fatigue, chest pain) can appear hours after a near-drowning event. Always seek medical review after a submersion incident.
- **CPR physiology:** Chest compressions create approximately 25–30% of normal cardiac output, delivering enough oxygenated blood to keep the brain alive until the heart can restart (spontaneously or via defibrillation).

Sources: St John Ambulance Vic — stjohnvic.com.au · Heart Foundation — heartfoundation.org.au · WHO Drowning Factsheet — who.int

Activity 2.1 — Physiology Research Task

Individual · ~15 min

Question 2.1a

3 marks

Explain the physiological sequence of events that occurs during drowning — from submersion to brain damage — in order. Use the terms: submersion, hypoxia, laryngospasm, cardiac arrest, cerebral hypoxia.

1.

2.

3.

4.

Question 2.1b

3 marks

Using the CPR physiology information above, explain why early bystander CPR is described as the single most important factor in survival from cardiac arrest caused by drowning. In your answer, refer to cardiac output, oxygenation, and the 4-minute window.

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Question 2.1c — Research 2 marks

What is secondary drowning and why is it clinically significant? Find one source that discusses it and write the URL below.

Secondary drowning is:

Why it matters clinically:

Source URL:

3 Social Determinants of Drowning Risk

KEY KNOWLEDGE — WHO IS AT RISK AND WHY

- Social determinants are the non-medical factors that influence health outcomes
- CALD communities may have limited swimming opportunity due to cultural norms, cost, or country-of-origin practices
- Rural and regional Victorians swim at unpatrolled inland waterways at higher rates than urban residents
- Alcohol is a factor in approximately 20–30% of adult drowning deaths in Victoria
- For drowning: gender norms, cultural background, socioeconomic status, and geography all play a role
- Masculinity norms ("don't back down from dares") are a documented risk factor for young male drowning
- People with lower socioeconomic status are less likely to have learned to swim as children

Sources: LSV Drowning Report 2023–24 · RLSSA National Drowning Report · WHO Social Determinants of Health Framework

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Activity 3.1 — Social Determinants Analysis

Pairs · ~18 min

Question 3.1a

4 marks

Choose TWO social determinants of drowning risk from the key knowledge above. For each, explain: (a) how it increases risk, and (b) what a realistic prevention strategy targeting that determinant would look like.

Social Determinant	How it increases risk	Prevention strategy
Determinant 1: *Write here...*		
Determinant 2: *Write here...*		

Question 3.1b

3 marks

Life Saving Victoria reports that 185 people from multicultural backgrounds fatally drowned in Victoria since 2015–16 — 37% of all drowning deaths. Propose THREE specific, practical actions a Victorian secondary school could take to reduce this disproportion — beyond just teaching the standard swim and water safety curriculum.

1.

2.

3.

4 Evaluating Drowning Prevention Strategies

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KEY KNOWLEDGE — PREVENTION APPROACHES AND EVIDENCE

- **Engineering:** Pool fencing (4-sided isolation) — strongest evidence base of any single intervention. Reduces child drowning in home pools by ~83%.
- **Education:** Swim lessons — strong evidence for reducing child drowning. Adults benefit less unless combined with behaviour change.
- **Technology:** AEDs, throw rescue equipment, CCTV at public pools, drowning detection systems.
- **What has actually worked:** fatal drownings involving alcohol and/or drugs are down **67%** against the Victorian Water Safety Strategy baseline, and pool fencing legislation cut drowning among children aged 0–4 by **63%** nationally. Both combined legislation with enforcement and public campaigns. Neither was education alone.
- **Supervision:** Lifeguard presence at patrolled beaches — Victoria recorded zero public-pool drowning deaths in 2024–25, the first time since 2019–20. Supervision cannot reach home environments, where 25% of deaths occurred.
- **Legislation:** Pool fencing laws, lifejacket requirements, alcohol restrictions near water.
- **Community programs:** LSV SwimSafe, RLSSA Swim and Survive, targeted multicultural programs.

Sources: WHO Drowning Prevention Guide — who.int · LSV Toolkit — lsv.com.au/toolkit · RLSSA — royallifesaving.com.au

Activity 4.1 — Prevention Strategy Evaluation

Individual · ~15 min

Evaluate the following prevention strategies using the framework: **Evidence strength · Target population · Limitations · Cost**

Strategy	Evidence strength (Strong / Moderate / Limited)	Limitations	Best target group
Pool fencing (4-sided)			
Mandatory swim lessons for primary school students			
Increasing patrolled beach patrols to 7 days / week			
Warning signs at unpatrolled beaches			

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Targeted CALD community swim programs			

Question 4.1a — Justification `4 marks`

If you were advising the Victorian Government on how to spend $5 million on drowning prevention, which TWO strategies would you prioritise, and why? Use evidence from your table and the key knowledge above to justify your recommendations.

Strategy 1:

Justification:

Strategy 2:

Justification:

5 International Approaches — Comparing Australia, UK, and USA

Activity 5.1 — Research Comparison Task `Individual / pair · ~20 min · Research required`

Visit the three organisations below. Research their approach to water safety education for secondary students. Complete the comparison table.

🇦🇺 RLSSA: royallifesaving.com.au | 🇬🇧 RLSS UK: rlss.org.uk/education | 🇺🇸 Red Cross: redcross.org/take-a-class/water-safety

Element	🇦🇺 RLSSA (Australia)	🇬🇧 RLSS UK	🇺🇸 American Red Cross

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Primary program name			
Key safety message / slogan			
Secondary school focus			
Emergency response approach			
Unique feature			

Question 5.1a `3 marks`

Based on your research, identify ONE element from an international approach that is NOT currently prominent in Victorian water safety education, and argue whether it should be adopted. Support your argument with evidence.

The element is: _____

Whether it should be adopted and why: _____

_____

_____

6 Diving, Spinal Injuries & High-Risk Teenage Behaviour

Key Knowledge — Australian Statistics & Research Context

Diving and jumping into unknown or shallow water is one of the leading causes of catastrophic, permanent injury among Australian teenagers — yet it remains heavily under-discussed in water safety education. The following data should inform both your understanding of the issue and your assessment task arguments.

10%
of all spinal cord injuries in Australia are caused by unsafe diving

20%
of quadriplegia cases in Australia result from diving injuries

95%
of aquatic spinal injury patients are male (major Australian review)

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75%+
of aquatic spinal injury patients are under 30 years of age

66%
of aquatic spinal injury patients suffer permanent disability

+34%
increase in drowning among 15–20 year olds post-COVID vs pre-COVID (RLSSA)

Sources: Royal Life Saving Australia (royallifesaving.com.au) · AIHW Diving Injury Report (aihw.gov.au) · Royal Perth Hospital spinal injury study

Highest-Risk Age Groups & Locations

The pattern of injury is not random — it is concentrated in specific age groups, behaviours, and locations. Understanding this pattern is essential for designing effective prevention.

Highest-risk age groups:

- Children aged 10–14 — highest incidence of diving-related spinal injuries
- 15–20 year olds — 34% increase in drowning post-COVID
- Males under 30 — more than 75% of all aquatic spinal injuries

High-risk locations:

- River bridges and wharves / jetties
- Dam walls and irrigation channels
- Rope swings over rivers and quarries
- Rock platforms and waterfalls
- Beaches with tidal sandbanks
- Floodways and drainage channels

Royal Life Saving Australia identifies **peer pressure, impulsive behaviour, alcohol, "showing off," jumping from structures, and overconfidence in swimming ability** as the major behavioural contributors to teenage aquatic injuries and fatalities.

Australian Case Studies — Recent Incidents

CASE STUDY 1 · NSW 2025

Terrigal Wharf — spinal injury

A teenage boy suffered serious spinal injuries after diving from a wharf at Terrigal into water he believed was deep. Helicopter evacuation to hospital. He was unaware of the reduced water depth beneath the wharf.

Key themes:

- Unknown depth beneath structures
- "It looked deep enough" misconception
- Hidden shallow zones under wharves

CASE STUDY 2 · VICTORIA 2025

Torquay Front Beach — surf beach diving

A teenage boy suffered a serious suspected spinal injury after a diving incident at Torquay Front Beach and was airlifted to hospital. Surf beach sandbanks shift constantly, making depth at any given point highly unpredictable.

Key themes:

- Sandbanks change with every tide
- Holiday risk-taking behaviour

CASE STUDY 3 · NSW 2024

Bawley Beach jetty — sandbank formation

A young man suffered serious injuries diving from a jetty near Bawley Beach. A sandbank had recently formed beneath the jumping area, dramatically reducing water depth. Prior experience at the location created false confidence.

Key themes:

- Depth changes — even at familiar sites
- "I've done it before" false confidence
- Tidal sediment movement

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- Catastrophic neck injury from one dive

- Lifelong consequences of one dive
- Surf beaches are not "safe" for diving

- Local knowledge ≠ current safety

Why Inland Waterways Are Australia's Highest-Risk Environment

Royal Life Saving Australia states: *"More people drown in inland waterways than any other location in Australia."* Five specific factors explain this:

1. Depth changes constantly

Drought, flooding, irrigation releases, sediment build-up, sandbank movement, and dam level changes all alter depth unpredictably — even between morning and afternoon.

2. Visibility is poor

Rocks, submerged trees, debris, logs, shopping trolleys, and submerged branches are frequently invisible beneath turbid (murky) water.

3. Cold water shock

Many Australian rivers and dams remain cold even in hot weather, triggering panic, muscle cramping, loss of coordination, and drowning within minutes.

4. Alcohol and group risk-taking

Inland water incidents frequently involve alcohol, teenage males, group peer pressure, and jumping from bridges or riverbanks — a combination that significantly multiplies risk.

5. Remote locations delay rescue

River and dam locations often mean delayed ambulance access, limited mobile phone reception, no lifeguards, and delayed spinal immobilisation — all of which worsen outcomes.

The core prevention message

"Feet first, first time." Never dive into unknown water. Always check depth every single time — not just the first time. Previous safety at a location does not guarantee current safety.

Activity 6.1 — Case Study Analysis

Using the three case studies above and the statistical data provided, answer the following questions. All responses should be evidence-based, using specific statistics where relevant.

6.1a

Identify two common factors present across all three Australian case studies. For each factor, explain the specific mechanism by which it contributed to the injury — not just that it was present.

6.1b

The Australian data shows that 95% of aquatic spinal injury patients are male. Using the social determinants framework from Section 3, construct a structural explanation for this statistic. Your answer must go beyond "males take more risks" and identify the social factors that drive this behaviour pattern.

6.1c

Two-thirds of Australian aquatic spinal injury patients suffer permanent disability. Why does a diving injury at the cervical (neck) spine carry a particularly high risk of permanent quadriplegia compared with injuries at other spinal levels? Use the physiological content from Section 2 to support your answer.

6.1d — Extension

The Bawley Beach case study illustrates the concept of *false confidence from prior experience*. Using this case as a foundation, argue why prior familiarity with a location may be **more dangerous** than complete unfamiliarity. What specific prevention message should follow from this argument?

Activity 6.2 — Myth vs Evidence Analysis

The following myths are extremely common among secondary-aged students. For each myth, write a clear evidence-based refutation — using specific Australian statistics or mechanisms where possible. The refutation must explain WHY the myth is false, not just assert that it is.

Common myth	Evidence-based refutation (write your response)
"If other people are jumping there, it must be safe."	
"Clear water means deep water."	
"I checked it before — I know this spot."	
"Good swimmers are safe from diving injuries."	
"It's safer to jump feet-first than to dive."	
"The jetty is high — so the water must be deep."	

Activity 6.3 — Prevention Strategy Design

Using the evidence from this section, design a targeted prevention strategy for one specific high-risk group and location. Your strategy must be grounded in the data above and must address the specific behavioural and structural factors driving risk for your chosen group.

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Target group

Identify your target group (e.g. "teenage males aged 15–18 at Victorian river swimming holes") and explain in one paragraph why this group faces disproportionate risk — citing at least two statistics.

Prevention strategy

Describe your strategy in specific terms: What will you do? Through what channel? Who delivers it? What behaviour change does it target? How will you know if it has worked (SMART metric)?

Limitation

Identify the most significant limitation of your strategy — and explain what a more complete response would look like if resources were not a constraint.

7 Assessment Task — Water Safety Policy Communication

🎯 Your Assessment Task

Using the research and analysis you have completed in this unit, produce ONE of the following. All options require evidence-based argument and must reference at least THREE credible sources with URLs.

Option A — Policy Brief (Extended Written Response)

Write a 600–800 word policy brief addressed to the Victorian Minister for Sport and Recreation, recommending THREE specific actions the government should take to reduce Victorian drowning deaths over the next 5 years. Must include current data, analysis of social determinants, and evidence-based prevention strategies.

Option B — Multimodal Campaign Product

Design a complete water safety awareness campaign targeting ONE specific high-risk group identified in Victorian drowning data (e.g. young males, CALD communities, inland waterway users). Must include: campaign name, visual concept, key message, evidence-based justification, target platform, and success metrics.

Option C — School Audit & Recommendation Report

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Conduct a water safety audit of your school — what does it currently do for water safety education? What is missing? Produce a structured report with recommendations, benchmarked against Victorian Curriculum 2.0 requirements and LSV best practice.

★ All options: minimum 3 sources · statistical evidence · clear audience · call to action or recommendation

Planning Space — Assessment Task

Option chosen:

Target audience / addressee:

Three sources I will use (with URLs):

1. _____

2. _____

3. _____

Key argument / central claim:

✏️ Unit Self-Assessment

Skill / Knowledge Area	Still developing	Solid	Can teach it
Calculate and interpret drowning rate data	○	○	○
Explain the physiology of drowning and CPR	○	○	○
Identify and analyse social determinants of drowning risk	○	○	○
Evaluate prevention strategies with evidence	○	○	○
Compare international approaches	○	○	○

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Produce an evidence-based policy argument	○	○	○

One question I still have:

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LSV: lsv.com.au · RLSSA: royallifesaving.com.au
WHO: who.int · Victorian Curriculum: victoriancurriculum.vcaa.vic.edu.au

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